SPATIAL ORIENTATION IN THE WHITE RAT*

BY H. C. WILCOXON
Yale University
AND
R. H. WATERS
College of Wooster

In a recent series of articles (3, 4, 2, 5) Tolman and his students have presented evidence in support of the theory of expectancy. According to this theory, when an animal learns a maze he learns a spatial orientation towards the goal and builds up an 'expectancy' of reward at that goal *place*. This theory is in opposition to the stimulus-response doctrine, which denies that any such mentalistic concept as expectancy need be introduced to explain behavior.

It is often difficult to distinguish between this theory of expectancy, or place learning, and the alternative stimulus-response theory, and it is even more difficult to design an experiment which will lend support to one of the theories and not to the other as well. There seems to be little doubt that an animal, after becoming familiar with a certain goal location as defined by certain differential cues at or near the goal location, can later exhibit a tendency to return to that location when started from a new position in the same environment. The recent series of articles on spatial learning from the California laboratory has served to illustrate this point clearly. These observations flatly contradict the older Watsonian chain-reflex theory of learning but by no means silence the newer and more flexible stimulus-response theories. As a consequence, it is believed that the legitimacy of the use of such expressions as 'expectancy' and 'spatial orientation' as terms descriptive of the factors underlying such behavior is still open to doubt.

No precise definition of spatial orientation is to be found in the studies reporting 'expectancy' behavior. Nevertheless it is clear that in order for the term to have legitimate descriptive status it must be defined in somewhat the following manner:

Spatial orientation refers to that condition within an organism which enables the organism to make a response to a *spatial relationship* of objects in the environment, which *relationship* is determined solely by the *relative positions* of the objects in the environment and the organism itself.[1]

* This study was carried out at the University of Arkansas.

[1] Thus spatial orientation as used here differs from directional orientation as used by Dashiell (1) in that the former depends upon a perception of a spatial relationship derived at least in part

It is to be noted that the 'spatial learning' in rats recently demonstrated by Tolman *et al.* has not been shown to be dependent upon the rat's perception of a spatial relationship of objects per se for the reason that in each of the four studies published there have been present in the environment certain differential cues to which the rats conceivably could have become conditioned. That is, the rats may have made their responses towards discriminable objects rather than as a result of any perceived spatial relationship of objects. Consequently, 'spatial learning,' as used by Tolman, is not necessarily dependent upon spatial orientation as defined above. It is believed, however, that the interpretation of this type of learning in terms of expectancy assumes the rat's ability to perceive such a relationship and thereby gain a spatial orientation in the sense we have defined it. Hence, for additional support of Tolman's theory it would be desirable to show:

1. That the rat is capable of building up a spatial orientation which is clearly dependent upon a *spatial relationship.*

2. That the rat will develop an expectancy of reward at the place towards which he becomes oriented.

3. That the rat will demonstrate this spatial orientation and expectancy by taking a short-cut route to the place. With these three things in mind the following experiments were performed.

EXPERIMENT I

A. Animals

The animals used in both of the present experiments were albino rats from a local stock and were between 75 and 90 days old at the outset of training. In Experiment I 32 animals, 16 male and 16 female, with no previous maze experience were employed. The food used was Purina Dog Chow checkers and the animals were run and fed between 7:00 and 10:00 P.M. in both experiments. Water was in the home cages at all times.

B. Apparatus

The apparatus used in Experiment I was Maze I, the plan of which is shown in Fig. 1. This maze consisted of pathways elevated 36 in. from the floor with all paths made of two-in.-wide pine painted flat black. The starting pathways were curved to the left in an arc of about 300 degrees and were each eight feet long. The straight center path running from the food platform at F-1 to the food platform at F-2 was also eight feet long. On this path, four in. from each end, were wire frames from which were suspended black curtains eight in. high and 10 in. wide. Short-cut pathways, each four feet long, could be attached to either side of either curved starting pathway, extending to the food platforms.

One checker of dog chow was on each food platform at all times with a wooden block placed behind the curtain at the food platform which was incorrect for a given run.

This apparatus was surrounded by a curtain of unbleached muslin hanging from a circular frame of heavy gauge wire suspended six feet from the floor. The top of the circular curtain

from exteroceptive stimulation, whereas the latter was believed to be dependent upon intra-organic cues which served to keep the organism oriented once it started moving in a given direction.

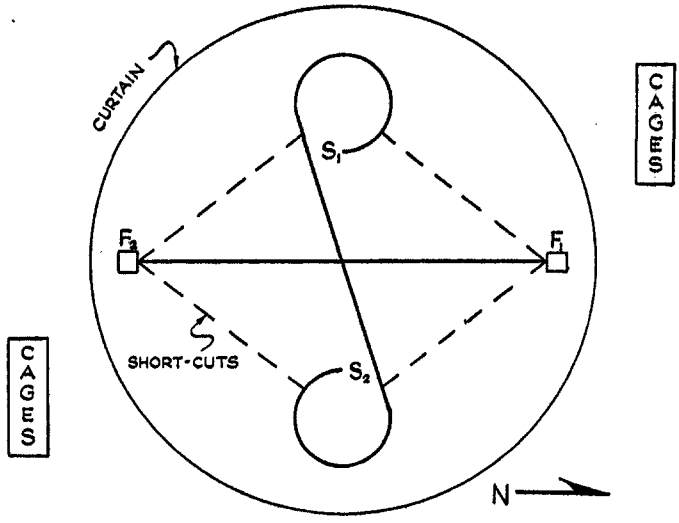


FIG. 1. This figure shows the floor plan of Maze I used in Experiment I. The dotted lines indicate the four possible short-cuts, one of which was added on the critical trial.

was covered with the same material and the floor of the room was painted flat black. The only illumination in the laboratory during the runs consisted of two shaded, 15-watt bulbs, one of which was placed directly over the center of the circle described by each starting pathway. This lighting arrangement made the curtain essentially a one-way vision screen. The entire apparatus was placed in a room 15 feet wide and 30 feet long, with the center pathway, F-1 F-2, running north and south.

The home cages were placed in positions symmetrical to the maze, as is shown in Fig. 1, since a previous investigation (unpublished) had indicated that auditory cues from the home cages would have to be controlled.[2] There were no other animals in the laboratory.

C. PROCEDURE

All animals were given identical training. Preliminary training was given on pathway F-1 F-2 on the first three days of the experiment. In this training the animals were fed on each of the two food platforms and given practice in running under the small black curtains to food.

On the fourth day the experimental training was begun with the curved starting pathways in position. Each animal was given four trials a day alternated in an ABBA order. Whenever an animal was started at S-1, the platform at F-2 was blocked. And conversely, when started at S-2, the platform at F-1 was blocked. Thus all of the animals were required to learn always to go left to obtain food. An error was recorded whenever a rat made a body length entry into either the pathway leading to the incorrect food platform or the pathway leading to the opposite starting position.

The training was continued until all of the animals either learned the problem to a criterion of 10 successive errorless runs, or developed stereotyped responses which seemed to indicate that they would never learn.

As the animals met the criterion of learning they were each given one critical trial on the maze with one of the short-cut pathways added. This critical trial was given on the same day the animal met the criterion. Half of the animals were tested to determine whether they would take a short-cut to the correct place, i.e., the platform at which they had been trained to 'expect'

[2] This confirms Ritchie's (2) finding that auditory cues from home cages are enough to support place learning.

food. These animals were designated the Experimental Group. The other half were tested to see whether they would take an identical short-cut pathway to the 'incorrect platform,' i.e., the platform at which they had never received food when started from the given position. These animals were designated the Control Group. Whether an animal was placed in the Control Group or in the Experimental Group depended upon when it met the learning criterion. The first animal to learn was placed in the Experimental Group, the second in the Control Group, the third in the Experimental, the fourth in the Control, etc. The starting position for the critical trial of any given animal was the same as it would have been on the next regular trial. When an animal in the Experimental Group was given its one critical trial, the only short-cut on the apparatus was the one leading from the given starting position to the correct place. Similarly, when an animal in the Control Group was given its one critical trial, the only short-cut on the apparatus was the one leading from the given starting position to the incorrect place.

Note that in each of the four possible short-cut setups an animal had to make a *right* turn off the starting pathway in order to get on the short-cut pathway, a response never rewarded in the training.

D. Results

Six of the animals failed to meet the criterion. All of these animals developed right-turning habits at the choice point, so that their performance consisted of at least one error on each trial after the habit became established.

The remaining 26 animals learned the maze to the criterion in an average of 7.9 trials, and half were placed in each group. Of the 13 animals placed in the Experimental Group, 12 took the short-cut presented to them, i.e., the short-cut to the correct food platform. Of the 13 animals placed in the Control Group, 10 took the short-cut presented to them, i.e., the one to the incorrect food platform. Of the four animals which learned the maze and were tested but did not take a short-cut, three proceeded without error over the old behavior route, and the fourth made a right-turning error at the old choice point.

E. Discussion

If significantly more animals had taken the short-cut to the correct place than to the incorrect place an interpretation in terms of spatial orientation would clearly have been indicated. It could have been argued that the animals had built up a spatial orientation which was dependent upon the spatial relationship of the starting position and the correct food platform, that they had learned to expect food at a place defined solely by this relationship, and that they had demonstrated these two facts by taking an entirely new pathway to the correct place and ignoring an identical one to the incorrect place. But almost equal numbers of animals took the two short-cuts, which renders the results equivocal for this fact is not evidence *against* the Tolman theory.

Consider the 26 animals which learned and were tested on the short-cuts. Twenty-two of them, or 85 percent, did take one of the short-cuts. In all cases, taking the short-cut involved the animal in

a right turn toward one of the food platforms, a response which had never been rewarded in the training. Three possibilities are offered in explanation of this result.

1. The animals might have taken the short-cut because it was a new stimulus which inhibited the old response and called forth some sort of exploratory behavior.

2. The animals might have taken the short-cut because they expected food behind the black curtain at the end of *either* short-cut pathway. However, if this were true it could be argued that the animals were responding to an object, i.e., a black curtain, and not to a spatial *relationship* of objects.

3. The animals might actually have built up a spatial orientation and expectancy of food at the correct place but did not demonstrate it clearly because of certain conflicting aspects of the experimental situation. The fact that on the critical trial each animal was faced with a new stimulus, viz., the short-cut, may have been important, as suggested above. And if a new stimulus could serve so effectively to inhibit the old response, it might also have inhibited any demonstration of spatial orientation and expectancy which, according to the Tolman theory, was an important factor in the old response. A better experiment would be one in which the short-cut, as a new stimulus, could not wield such a decisive influence. An apparatus which would allow the simultaneous presentation of two short-cuts, one to the correct place and the other to the incorrect place, would overcome this difficulty.

Each of the above possibilities appeared equally likely and suggested further research.

EXPERIMENT II

A. Introduction

In the preceding experiment the evidence indicated that the short-cut pathways, acting as new stimuli, might have elicited exploratory, rather than expectancy behavior. The present experiment was designed to overcome this difficulty. An apparatus was designed which allowed the simultaneous presentation of two short-cuts, so that the animal was required to make a choice between two new pathways, rather than between an old one and a new one as in the previous experiment. Thus any tendency toward exploration was automatically equalized as regards the pathway the animal might take on the critical trial.

B. Animals

Ten male rats with no previous maze experience were used.

C. Apparatus

The apparatus was the same elevated maze described in Experiment I with the exception that straight starting pathways 30 in. long were substituted for the curved ones used in the previous experiment (see Fig. 2).[3]

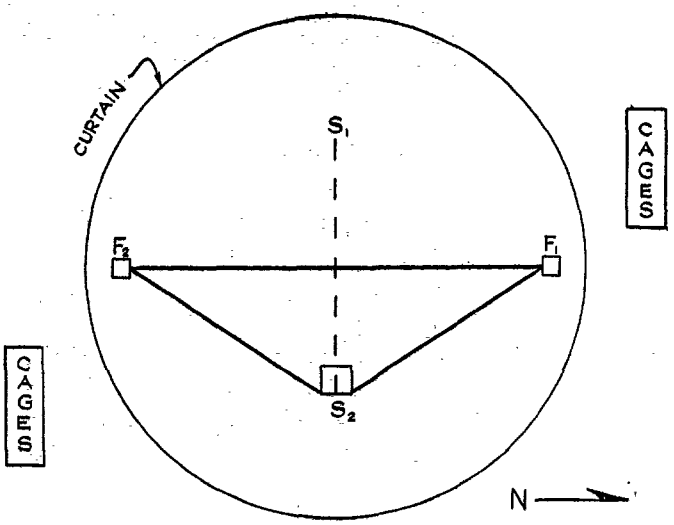


Fig. 2. Floor plan of the maze used in Experiment II. A typical short-cut setup. The dotted lines indicate the straight starting pathways which were removed when the short-cut tests were given.

The short-cuts which were added to this apparatus on the critical trials at the termination of the experimental training were two-inch wide paths similar to pathway F-1 F-2. They were attached to a platform eight in. square which was placed at one of the starting positions and extended in opposite directions to each of the two food platforms. The regular starting pathways were removed for the critical test trials. The apparatus was placed inside the circular curtain in the same position occupied by the maze in the previous experiment. The home cages of the animals, which were the only ones in the laboratory at the time, were placed symmetrically about the apparatus.

D. Procedure

Five days before the experimental training the animals were placed on a 24-hour maintenance schedule. On the two days preceding the experimental training the animals were given preliminary training on pathway F-1 F-2, which consisted in allowing each animal to eat for five min. on each of the two food platforms. On the final night of preliminary training the animals were also allowed to run down the pathway to food five times.

In the experimental training the animals were trained always to go either to the right or to the left at the choice point. A preliminary trial with no food on either platform was given each animal to determine his initial turning preference. On all subsequent trials he was trained against this initial preference. As a result of this test eight animals were trained always to go to the right and two were trained always to go to the left.

Each animal was given four trials a day with starting positions rotated in an ABBA order. If an animal was being trained to go always to the right and was started from S-1 he found food at F-2, and conversely, if started from S-2, he found food at F-1. The procedure for an

[3] The small black curtains in front of the food platforms were removed before this experiment was begun since the visual acuity of the white rat was not considered high enough to enable it to see a small pellet of food four feet away.

animal being trained to go always to the left was the converse. This training was continued for 12 days.

As an animal met the criterion of 10 successive errorless runs the straight starting pathways were removed and the short-cut pathways put in on the side of the next regular starting position. Food was placed on both platforms on all the short-cut test runs. The animal was then brought from his home cage and placed on the eight-in.-square platform facing pathway F-1 F-2 and released. On the following day he was given one trial on the double-T maze with food only at the correct platform followed by another test on the short-cut with food at both places. This procedure was followed until the experiment was discontinued on the 12th day.

E. Results

Three of the 10 animals failed to learn within 48 trials to go always to the right (or left) to obtain food. Instead they developed stereotyped habits of turning always in the wrong direction at the choice point, running to the empty food platform, and then to the correct one. The remaining seven rats met the criterion in an average of 11.1 trials. When these animals were tested on the short-cut immediately after meeting the criterion, four took the short-cut to the correct place, i.e., the one to which they had been trained to go from the given starting position, and three took the short-cut to the incorrect place.

The results obtained on the following days when the animals were given both a reinforcing trial on the double-T and a test trial with the short-cuts in place indicated that the animals made no improvement in going to the correct place over the short-cuts, and that individual rats were inconsistent in their choices during the first few days of the tests. As the tests continued from day to day the animals became consistent in their choices of short-cut, but these choices seemed to bear no relation to the previous training.

It is concluded that the animals showed no spatial-orientational ability in this situation. It should be noted that this fact cannot be taken as proof that the rat is incapable of acquiring a spatial orientation, even in the precise sense in which it has been defined, but only that our rats in our situation did not exhibit it.

Summary

1. Spatial orientation is defined in the present paper as that condition within an organism which enables the organism to make a response to a spatial relationship of objects in the environment, which relationship is determined solely by the relative positions of the objects in the environment and the organism itself. The two experiments reported here were performed in an attempt to determine whether the white rat is capable of such spatial orientation.

2. In Experiment I 32 rats were trained to go always to the left on a modified double-T elevated maze with curved starting pathways.

This maze was placed in an environment which was homogeneous with regard to visual and auditory extra-maze cues. Twenty-six animals learned the criterion (10 successive errorless runs) within 48 trials and were each given one critical trial with a short-cut added. The Experimental Group ($N = 13$) were tested with a short-cut leading to the correct place. The Control Group ($N = 13$) were tested with a short-cut leading to the incorrect place. Nearly all of the animals in both groups took the short-cut regardless of whether it led to the correct place or the incorrect place, and in spite of the fact that taking it involved a right turn never before rewarded. This result is interpreted as equivocal, neither affirming nor denying the rat's ability to gain a spatial orientation.

3. In Experiment II 10 rats were trained to go always to the right (or left) on a conventional double-T elevated maze placed in the same homogeneous environment as that of Experiment I. Seven of the rats learned within 48 trials. Each of these rats was then presented two short-cuts simultaneously, one leading to the correct place and one to the incorrect place. No preference was exhibited for the correct short-cut, even though the tests were continued for several days after the animals first met the criterion.

4. It is concluded that the negative results reported here do not prove that the rat is incapable of acquiring a spatial orientation, but only that the white rats used in these experiments exhibited no evidence of having acquired a spatial orientation in the sense in which the term was defined.

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